

CHronoZ

MACRO INTELLIGENCE AGENT

Empirical Benchmark Research Layer, Relationship Maps & Agent Integration

A research update that turns the completed CHronoZ economic-benchmark notebook program into a raw, auditable historical-normal reference layer for human and LLM macro reasoning.

CANONICAL RESEARCH UPDATE - v0.3

This version updates v0.2 after the repository-wide empirical study: matrix correlation, rolling correlation, lead-lag, redundancy, native distributions, Delta Distribution audits, and alternative benchmark families. Macro categories now organize evidence rather than forcing one composite graph or one total state score.

QuantFun x Wonnyy / 20 September 2026

WHITEPAPER v0.3

RESEARCH UPDATE

01. Abstract

CHronoZ is evolving from a conceptual macro-state engine into an empirical benchmark research layer. The repository now contains 36 cleaned U.S. macroeconomic series, 44 executed or auditable Jupyter notebooks, relationship studies across transformed monthly indicators, full native-frequency distribution inspection, a 71-candidate Delta Distribution study, and dedicated audits for difficult series. The research objective is deliberately modest: estimate what has historically been ordinary or unusual for each transformed variable, preserve its context, and expose relationships without pretending that correlation is causality or that a benchmark is a forecast.

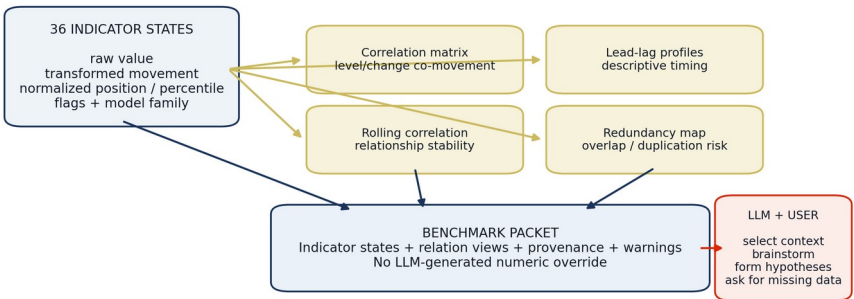
The most important architectural change from v0.2 is that macro categories such as Growth, Labor, Inflation, Credit and Liquidity are no longer assumed to collapse into one canonical node score or one chart. They remain useful organizational tags, but the canonical quantitative product is a collection of indicator-level benchmark states plus relationship views: correlation, rolling correlation, lead-lag profiles, redundancy screens and data-quality metadata. A human researcher or LLM can then decide which functions, contrasts and additional data matter for the question at hand.

Plain-language thesis

The benchmark is not the answer. It is the coordinate system. It tells the researcher how unusual a movement is relative to its own history and which other variables have historically moved with, against, before, after, or redundantly beside it. Interpretation remains a research task.

Benchmark architecture: preserve the indicators, surround them with relationship views

Macro categories organize retrieval; they do not require a single composite score or a single graph.



Interpretation quality remains downstream: the benchmark supplies measurable coordinates; the user and LLM decide what questions those coordinates make worth investigating.

Figure 1. The canonical v0.3 benchmark architecture: indicator states remain visible; relationship maps surround them; the LLM and end user reason on top.

What changed from v0.2

Earlier framing	Updated framing
Macro nodes may become composite scores	Categories are primarily indexing/retrieval labels; aggregation is optional and must be separately validated.
State vector is the preferred core object	Indicator-level benchmark states + relationship matrices are the canonical measurement substrate.
One normalization family is the default	Series can use selected delta, robust delta, discrete PMF, regime-conditioned robust scoring, or structural-epoch

Earlier framing	Updated framing
	scoring.
Correlation supports node design	Correlation, rolling paths, lead-lag and redundancy are independent research views, not an automatic graph.
Benchmark output points toward interpretation	Benchmark intentionally remains raw; end-user context and LLM reasoning determine what question to investigate.
Backtest follows after model construction	Validation now explicitly includes benchmark stability, relationship stability, agent usefulness and cross-asset mapping as separate tests.

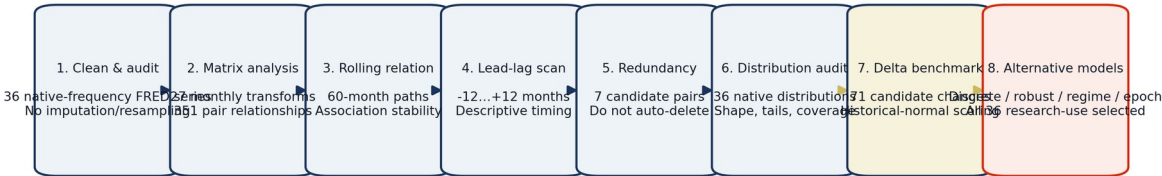
EMPIRICAL BASE

02. Research Repository: What Exists Now

The current repository is not a trained macro model. It is a research laboratory with a strong data-audit layer and a growing set of statistical notebooks. The active cleaning entry point is dataCleansing.py; it scans raw FRED CSVs, preserves raw files, removes only clearly defined invalid or missing-value rows from cleaned training inputs, and does not fill, interpolate, resample, normalize or economically transform the data. The older src/chronoz_quant modules remain in the repository as legacy work and should not be treated as the active production pipeline.

CHronoZ empirical benchmark research program

The repository evolved from data integrity and relationship mapping toward indicator-specific historical-normal benchmarks.



Key research rule Relationship studies describe structure; delta/alternative models define benchmark scoring; neither alone proves causality or predictability.

Figure 2. Repository research sequence from data integrity to a model-family-aware historical benchmark layer.

Current quantitative universe

Category	Series count	Role in the research universe
Growth / Activity	8	Production, consumption, income, retail and housing activity
Labor	8	Payrolls, unemployment, claims, hours, openings, participation and wages
Inflation	6	Consumer, PCE and producer price pressure
Credit	5	Loan quantities, lending standards/demand and delinquency
Policy / Liquidity	5	Policy rate, reserves, central-bank balance sheet and M2
External / Structural	4	Trade, productivity and capacity utilization

All 36 series currently have a research-use benchmark selection: 30 are approved under the current delta/change approach, while six use alternative, regime-conditioned or structural model families. This is still a research selection rather than proof of stationarity, predictive power or production readiness.

RESEARCH INTERPRETATION

03. From Matrix Analysis to Benchmark Modelling

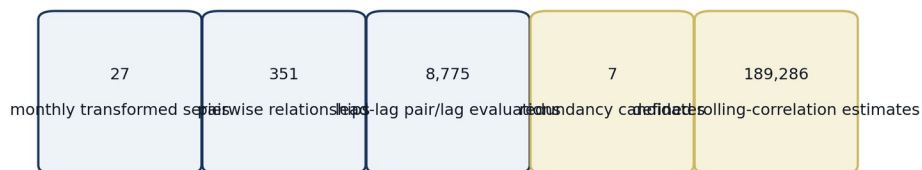
The empirical work did not begin by asking for one economic score. It began by asking whether the series behave as a system at all. The matrix notebook transformed the monthly subset into comparable one-month changes and inspected 351 pairwise relationships. That was useful, but it also exposed the first design constraint: a relationship matrix is not a definition of normal. Two series can be highly correlated and still have very different distributions, tails, timing properties and economic meanings.

The next notebooks therefore attacked the same data from different angles. Rolling correlation asked whether a full-history relationship survives through time. Lead-lag analysis asked whether the association is strongest at the same month or at another alignment. Redundancy screening asked whether apparently separate indicators may be telling almost the same story. Native distribution notebooks then moved back to the individual variable and made its shape, coverage and tails visible. Finally, the Delta Distribution study asked the question that directly matters for a benchmark: after an economically defensible transformation, can the history of changes serve as a reference distribution for what is ordinary and what is unusual?

Interpretive consequence

Matrix analysis is a map of relationships. Delta/alternative modelling is a map of historical normality. They should be delivered together to the agent, but they answer different questions.

Relationship-study snapshot



Strongest full-history transformed Pearson: INDPRO vs TCU = +0.9837. Most negative: PAYEMS vs UNRATE = -0.8786.

These are descriptive associations. Lead-lag peaks, rolling changes and redundancy screens are context for research, not causal or predictive claims.

Figure 3. Relationship-study snapshot from the current repository outputs.

RELATIONSHIP VIEW I

04. Matrix Correlation Analysis

The matrix-corrAnalysis notebook currently uses 27 monthly series and explicit one-month changes. Weekly and quarterly series are excluded rather than silently resampled. Each pair requires at least 60 overlapping observations; the actual transformed overlaps range from 199 to 1,290 observations. Pearson and Spearman are both retained because linear and rank association can tell different stories.

Finding	Current result	Interpretation
Monthly transformed series	27	Relationship analysis is frequency-specific; nine weekly/quarterly series are intentionally outside this matrix.
Unique pairs	351	Every monthly pair has defined Pearson and Spearman coefficients.
Highest positive Pearson	INDPRO vs TCU: +0.9837	Industrial production and capacity utilization changes overlap extremely strongly in the full sample.
Most negative Pearson	PAYEMS vs UNRATE: -0.8786	Payroll and unemployment changes move strongly in opposite directions in the full sample.
Minimum overlap	199 observations	The matrix is pairwise rather than listwise; histories are not artificially shortened to one common start date.

A critical v0.3 distinction is that this matrix transformation registry is an exploratory relationship layer, not the final benchmark registry. Later Delta Distribution work refined many preferred transforms toward log changes or alternative model families. The matrix remains useful as a structural reference, but its transformations must not silently override the final benchmark model selection.

RELATIONSHIP VIEW II

05. Rolling Correlation

Full-history correlation can hide relationships that reverse sign or change strength. The rolling-correlation notebook therefore computes trailing 60-calendar-month Pearson paths with at least 48 paired observations and requires the full elapsed window. The current run contains 189,286 defined rolling estimates across 351 pair paths, and all 351 pairs are defined in the latest 2021-09 to 2026-08 window.

Some relationships are visibly unstable. For example, real income excluding transfers versus total real personal income spans roughly -0.81 to +1.00 across rolling windows, while PCEC96 versus M2 spans roughly -0.96 to +0.69. This is exactly why CHronoZ should surface relationship stability rather than store a single eternal coefficient.

Rule

A changing rolling correlation is evidence that the association is state-dependent or sample-dependent. It is not, by itself, evidence of a hidden regime or a causal break.

RELATIONSHIP VIEW III

06. Lead-Lag Analysis

The lead-lag notebook evaluates every monthly pair at lags from -12 to +12 months, producing 8,775 pair-lag evaluations. The lag definition is $\text{corr}(X_t, Y_{(t+k)})$; a positive k means the first series is earlier in the aligned observations. Of 351 pairs, 194 have their peak absolute Pearson correlation at zero lag and 157 peak away from zero.

The nonzero peaks are useful as research prompts, not predictive claims. The largest gains over zero lag are concentrated in several real-income relationships, and some one-month timing patterns also appear in labor and money variables. But searching 25 lags for every pair creates a large multiple-comparison

problem, and the notebook deliberately does not incorporate publication timing. Before any lead-lag relationship becomes an agent rule, it must survive release-aware, walk-forward testing.

Agent use

Lead-lag output should be queried as a hypothesis generator: “this pair historically aligns more strongly with X earlier than Y.” The LLM must still inspect the economic mechanism, release timing and stability before treating that timing as meaningful.

RELATIONSHIP VIEW IV

07. Redundancy Mapping

Redundancy analysis uses a deliberately strict screen: at least 60 shared changes, absolute Pearson of at least 0.80, absolute Spearman of at least 0.60 and matching signs. Seven candidate pairs pass. Ten monthly series participate in at least one candidate connection; seventeen do not.

Candidate pair	Pearson	Spearman	Why it matters
INDPRO / TCU	0.984	0.963	Potential duplication of industrial-cycle information across category labels.
INDPRO / IPMAN	0.956	0.896	Broad production and manufacturing production are close but not automatically interchangeable.
IPMAN / TCU	0.938	0.856	Another industrial-capacity overlap to account for in interpretation.
PCEC96 / RRSFS	0.921	0.688	Consumption and retail activity can carry overlapping demand information.
CPI / PCEPI	0.911	0.883	Headline consumer inflation measures overlap strongly.
PCEPI / Core PCE	0.831	0.812	Price-level family overlap needs explicit treatment if any aggregation is later attempted.
PPIFID / PPIACO	0.815	0.779	Producer price measures overlap but retain different upstream/final-demand meanings.

The output is a warning against double counting, not an instruction to delete one side of every pair. In v0.3, redundancy becomes metadata for the agent and a constraint for any future composite view. If two indicators are highly redundant, an LLM should not present their agreement as two independent confirmations without qualification.

INDIVIDUAL BEHAVIOR

08. Distribution Analysis: Before and After Transformation

The 36 data-learning notebooks intentionally plot cleaned native levels with no transformation, imputation, trimming or model fitting. Their role is diagnostic: expose scale, trend, discrete steps, tails, gaps and coverage before any benchmark model is chosen. This native-level view is not the benchmark itself. A price index that drifts from 20 to 300 over decades can have a visually coherent histogram while still being an invalid historical reference for current price pressure.

The Delta Distribution study therefore evaluates change representations at native frequency and skips changes across calendar gaps. Across 36 series it evaluates 71 candidate change distributions. The final research selection is variable-specific because “change” is not one universal operation: positive quantities often favor log differences, rates and survey balances favor arithmetic changes, discrete rounded series may require probability masses, and policy-liquidity series may require regime conditioning.

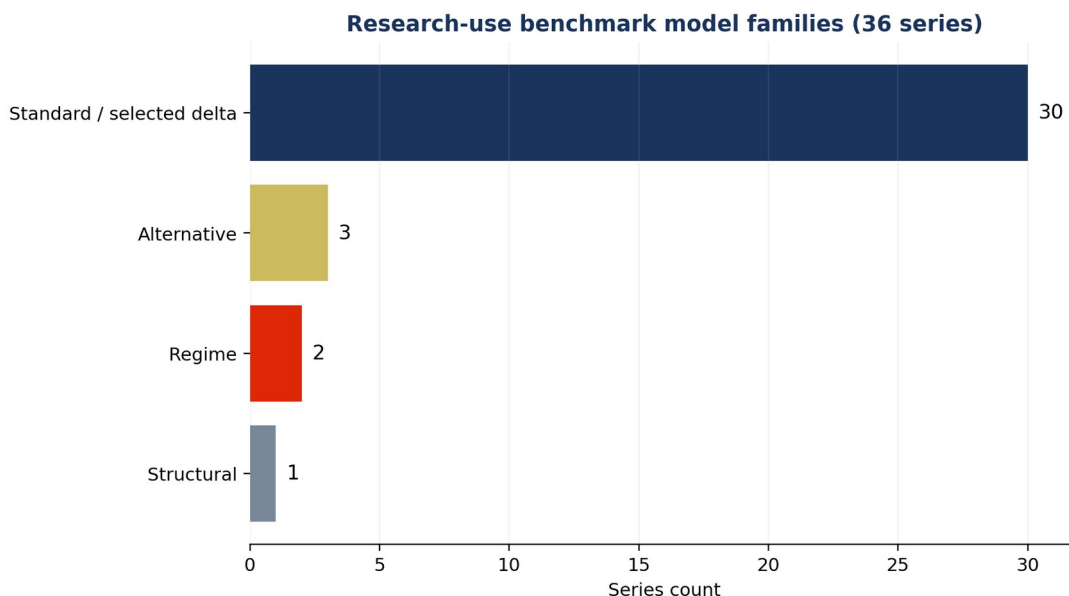


Figure 4. Research-use model-family selection after manual audits. The 30-series group includes the selected current delta approach; six series explicitly use alternative/regime/structural families.

CORE MEASUREMENT IDEA

09. Delta Distribution as the Historical-Normal Benchmark

For the majority of series, the most useful raw benchmark question is not “what is the average level?” but “how unusual is the latest transformed movement relative to historical movements of the same variable?” This is the Delta Distribution idea. For a positive trending quantity, one common transform is $g_t = 100 * \ln(X_t / X_{(t-1)})$. For a rate, arithmetic change $X_t - X_{(t-1)}$ is usually more interpretable. The historical distribution of that transformed movement becomes the reference coordinate system.

```
standard score: z_t = (delta_t - mean_reference) / std_reference
robust score:  z*_t = (delta_t - median_reference) / (1.4826 * MAD_reference)
empirical rank: p_t = count(reference <= delta_t) / N
```

The normalized value is not required to be Gaussian. The z-score is a unit of distance from a chosen center and scale; the empirical percentile is a distribution-shape-aware companion. The useful output is therefore

richer than one number: transformed value, center, scale, percentile, range label, direction, tail/outlier flag, model family, reference cutoff and provenance.

“Normal historically”

Normal means ordinary relative to the chosen historical reference process. It does not mean healthy, desirable, causal, investable or likely to persist.

WHY ONE TAIL POLICY CANNOT FIT EVERY SERIES

10. Two Case Studies: PPIACO and PCEC96

PPIACO - retain tails, normalize the movement

PPI All Commodities is a useful example of a sensitive indicator whose tails can be economically informative. The selected transform is one-month log growth. The dedicated audit keeps all tail observations and uses a standard z-score plus empirical percentile, while explicitly retaining regime-sensitivity warnings. An extreme reading is not automatically “inflation” or “stagflation”; it is a statistical prompt to cross-check producer prices, consumer inflation, industrial activity and any commodity information supplied by the end user.

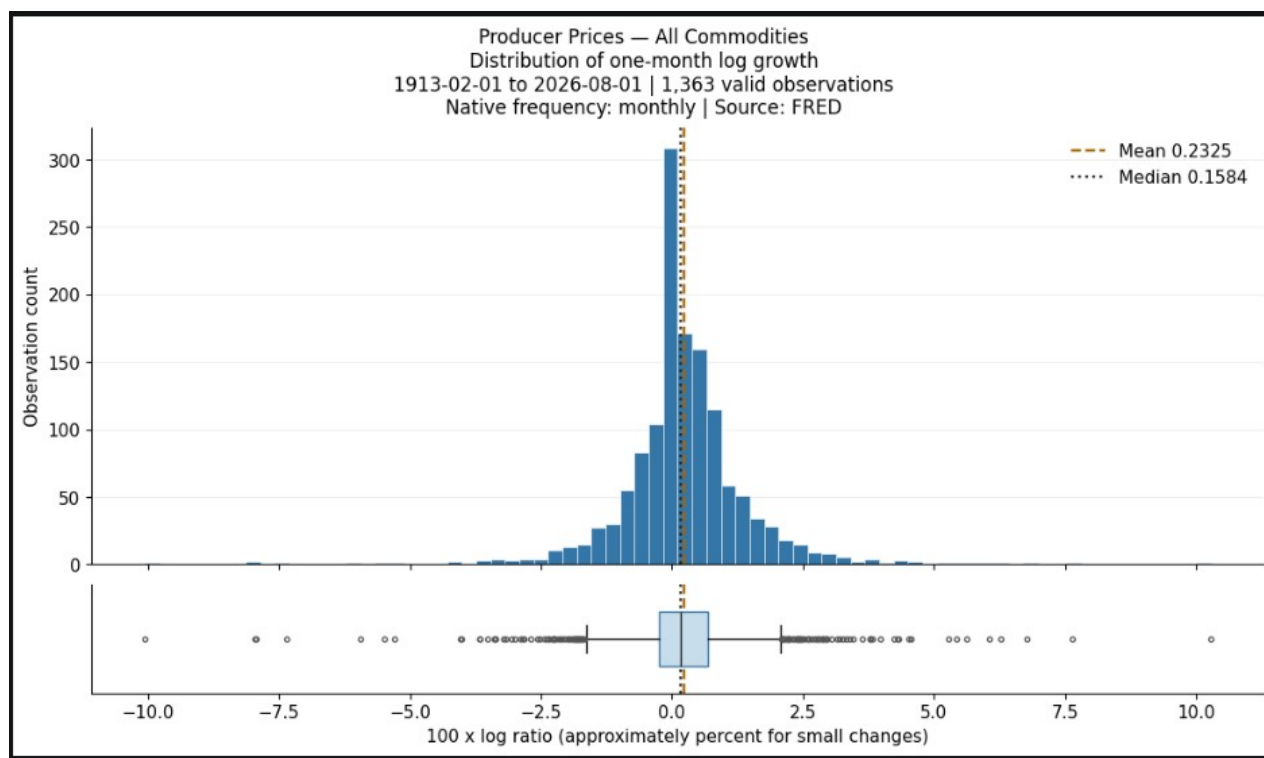


Figure 5. PPIACO one-month log-growth distribution used in the dedicated audit. Long tails remain part of the historical record.

PCEC96 - preserve the events, remove flagged disruptions from ordinary calibration

Real consumption shows a different pathology. The bulk of monthly log growth is tight, but a small number of shutdown/reopening observations dominate standard deviation and kurtosis. The dedicated audit therefore preserves every event but uses median/MAD modified-z screening to exclude flagged

observations from ordinary benchmark calibration. The excluded events are still scored against the benchmark and remain visible as historical disruptions.

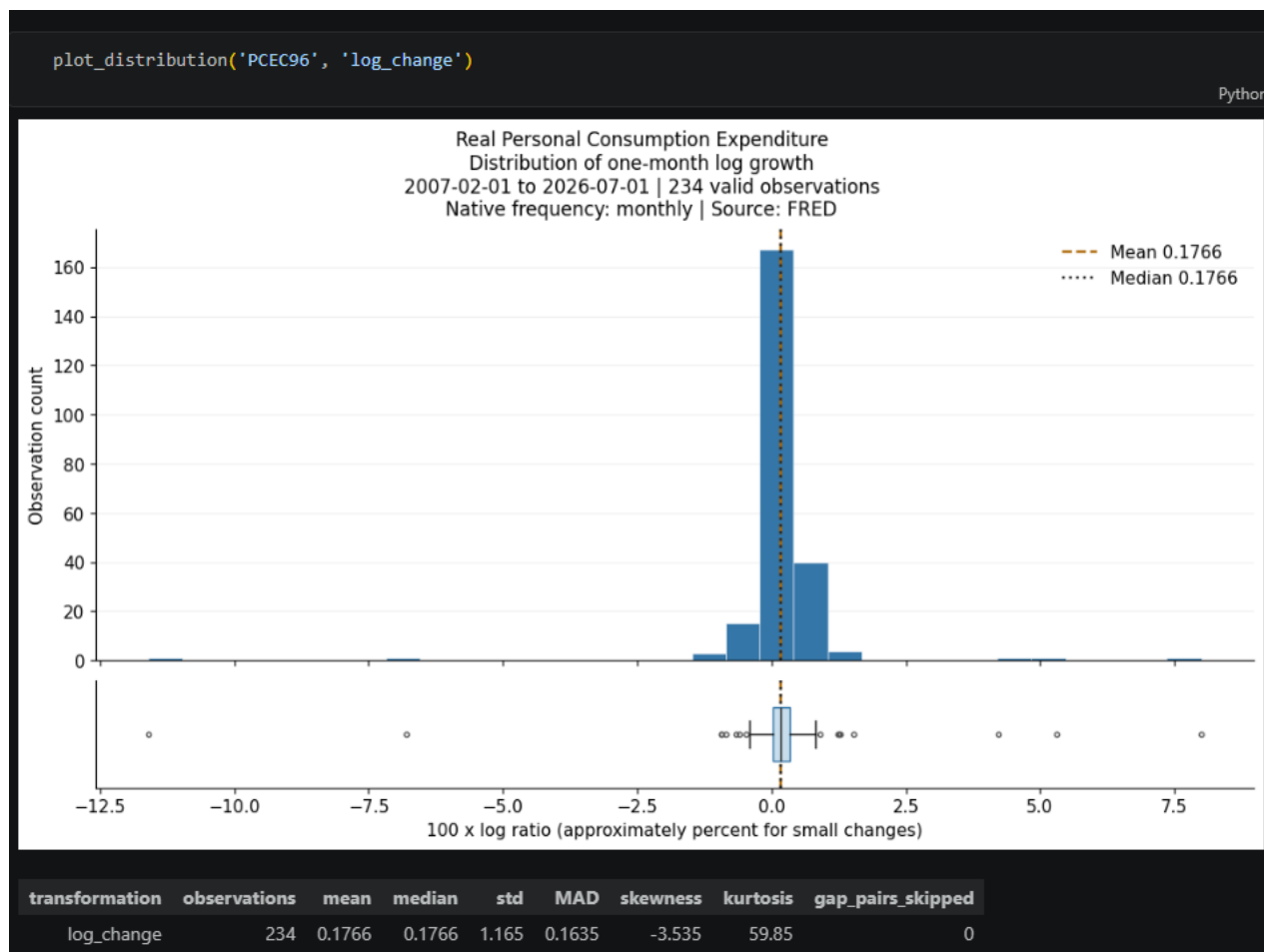


Figure 6. PCEC96 log-growth distribution. The central process is narrow relative to the COVID-era tails.

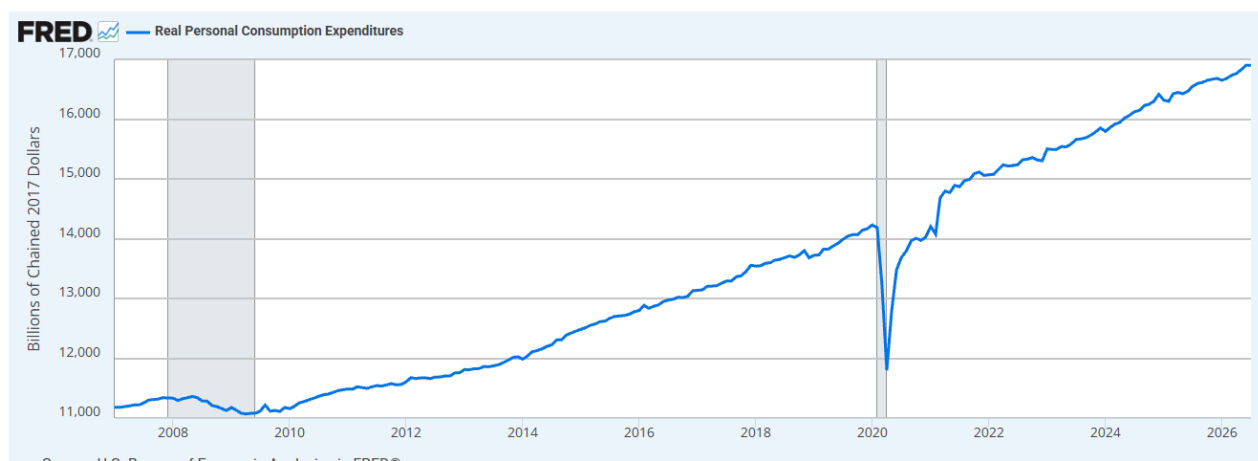


Figure 7. PCEC96 level history makes the 2020 shutdown/reopening event visually explicit.

In the current audit, 12 of 233 calibration-eligible months are excluded by the statistical rule, leaving 221 ordinary calibration observations. Standard deviation falls from about 1.1672 to 0.2717 and excess kurtosis from about 59.6 to 0.43. The value of this example is methodological: the benchmark can treat an economically real observation as “not representative of ordinary calibration” without deleting it from history.

NON-STANDARD SERIES

11. Alternative Benchmark Families

Six series were previously poor fits for one unconditional continuous distribution. The non-standard audit reframed the problem: keep the variables, but use a model family closer to the way they actually move. The result is not six bespoke black boxes; it is a small family of transparent alternatives.

Series	Selected family	Core idea	Current caution
AWHAETP	DISCRETE_DELTA	Empirical PMF / frequency of discrete hour changes	About 48% zero changes; exact-step probabilities depend on measurement precision.
ICSA	ROBUST_DELTA	Median + MAD + empirical percentile; retain crisis tails	Claims are inherently crisis-sensitive; robust scoring should not erase that property.
UNRATE	DISCRETE_PLUS_SHOCK	Discrete ordinary changes plus a robust shock detector	Shock threshold and discrete-state design require further walk-forward validation.
WALCL	REGIME_ROBUST_DELTA	Within-regime median/MAD scoring of weekly log growth	Current speed buckets are statistical states, not automatically QE/QT labels.
WRESBAL	REGIME_ROBUST_DELTA	Regime-conditioned robust liquidity scoring	Operational liquidity plumbing creates unusually unstable variance.
TOTRESNS	STRUCTURAL_EPOCH + ROBUST_DELTA	Segment structural monetary epochs, then score robustly inside epoch	Epoch boundaries are hypotheses that still need validation.

The audit router independently points in the same direction for most of these series. It exactly matches AWWHAETP, ICSA, UNRATE and WRESBAL, upgrades WALCL from regime delta to regime-robust delta, and identifies TOTRESNS as both heavy-tailed and structurally unstable. This agreement is encouraging, but the router remains a development diagnostic rather than a production model selector.

CORE DESIGN

12. The v0.3 Paradigm: A Multi-View Benchmark, Not One Macro Graph

The repository now suggests a different definition of “economic state” from the one used in v0.2. A single Growth or Inflation score may eventually be useful for a dashboard, but it should not be the canonical truth object. The canonical object should preserve the individual benchmarked indicators and the statistical views that describe their relationships. Categories remain useful because they tell the agent where to look; they do not require arithmetic aggregation.

This is closer to a research terminal than an index. Industrial production can be unusually weak, housing can be ordinary, retail can accelerate and real income can behave differently at the same time. Collapsing those signals too early destroys the very disagreement that often creates a research question. Likewise, PPIACO can be extremely elevated while industrial output is ordinary. The contrast is not a modelling error; it is the information gap that should trigger more research.

Canonical quantitative object

BenchmarkPacket = {indicator states, relation views, data-quality/provenance, model registry}. Optional composite nodes can be added later as derived views, never as replacements for the underlying evidence.

WHAT THE AGENT SHOULD RECEIVE

13. Benchmark Output Contract

The integration contract should be machine-readable and deliberately boring. It should give an LLM enough metadata to reason without asking the LLM to reconstruct the statistics. Different model families may expose different primary scores, but the schema should still be consistent.

```
BenchmarkPacket
  as_of
  economy
  benchmark_version
  data_vintage
  indicator_states: { ... 36 series ... }
  relation_views:
    correlation
    rolling_correlation
    lead_lag
    redundancy
  warnings
  provenance

IndicatorState
  series_id / category / frequency
  raw_value / observation_date
  transform / transformed_value
  model_family / score_type / score
  empirical_percentile / range_label
  direction / tail_flag / outlier_flag / regime_or_epoch
  reference_cutoff / reference_n
  data_quality / provenance
```

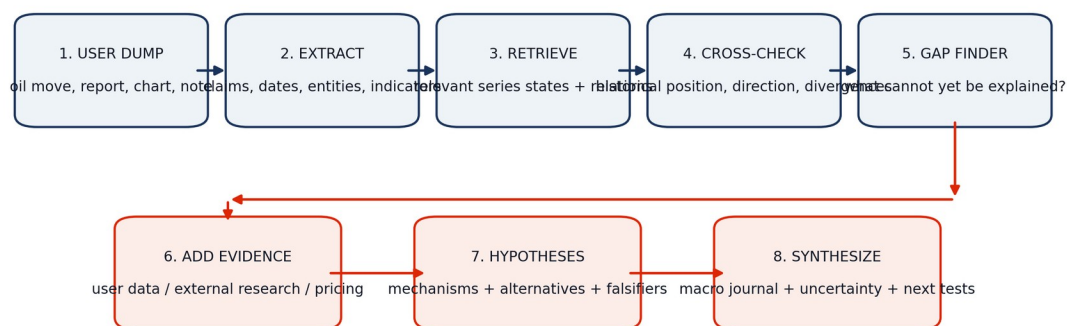
For a discrete series, score_type may be “empirical_probability” or “rarity” rather than z-score. For a regime-conditioned series, the reference metadata must identify the selected regime. For PCEC96, the packet should distinguish “full-history observation” from “included in ordinary calibration.” The agent should never infer these distinctions from prose alone.

WORKFLOW

14. How the Agent Should Analyze the Benchmark

How CHronoZ should reason with the benchmark

The quantitative layer answers “how unusual?”; the reasoning layer asks “why, what conflicts, and what evidence is missing?”



Hard boundary The agent may query and explain benchmark outputs, but it must not rewrite a numeric benchmark because a narrative sounds persuasive.

Figure 8. Proposed CHronoZ reasoning workflow on top of the quantitative benchmark packet.

A CHronoZ run should begin with the user’s research question or dump, not with a generic attempt to narrate all 36 indicators. The extraction layer identifies claims, entities, dates, markets and candidate economic variables. The agent then requests only the relevant benchmark states and their nearest relationship views. If the user drops oil and commodity data, for example, the agent may retrieve PPIACO, PPIFID, CPI/PCE families, industrial production, manufacturing production and capacity utilization; it does not need to summarize every labor and credit variable unless the evidence points there.

The next task is contrast, not conclusion. The agent asks: Is the latest movement central, elevated, extreme or a flagged disruption relative to the correct reference? Do related indicators agree? Is the relationship historically stable? Are apparently independent confirmations actually redundant? Does a lead-lag profile suggest another variable worth checking? If the configuration is internally coherent, the agent can summarize it. If it is contradictory, the contradiction becomes a research prompt and the agent should ask for or retrieve additional evidence.

Illustrative reasoning pattern

User evidence: oil +15%, copper +8%, iron ore +6%

Benchmark: PPIACO extreme positive; PPIFID elevated

Growth checks: INDPRO near normal; IPMAN slightly weak; TCU near normal

Relationship context: producer prices correlate with inflation measures, but not enough to prove pass-through

Agent conclusion: upstream price pressure is unusually strong relative to current industrial acceleration

Next question: supply disruption, foreign demand, inventories, freight, FX, policy, or another omitted factor?

Notice what the agent did not do: it did not convert an extreme PPI reading into a predetermined stagflation label, and it did not average inflation and growth into one score. The benchmark identified the unusual geometry; reasoning and new evidence attempted to explain it.

HUMAN AGENCY

15. User and LLM as the Brainstorming Layer

v0.3 intentionally keeps the benchmark “raw.” That is not a weakness; it is a design choice. Macro interpretation is path-dependent, context-heavy and often underdetermined by the public data alone. An end user may bring shipping rates, commodity curves, company commentary, positioning, a central-bank speech or an industry report that the base dataset does not contain. The LLM is useful because it can map this unstructured evidence onto the measured benchmark, propose mechanisms, identify contradictions and suggest what is missing.

This makes the LLM a research partner rather than the source of numeric truth. The quality of the final output depends on the evidence available, the questions asked, the discipline of the reasoning loop and the user’s willingness to challenge the narrative. CHronoZ should therefore preserve alternative explanations and falsifiers, not just produce a confident paragraph.

Division of labor

Python/statistics: calculate. Repository: remember. LLM: connect and question. User: decide what matters and judge whether the explanation survives contact with reality.

MARKET VALIDATION

16. Cross-Asset Research: Downstream, Not Inside the Benchmark

Cross-asset work should be integrated after the real-economy benchmark is stable enough to audit. Rates, FX, credit, equities, commodities, gold and volatility can then be normalized relative to their own histories and compared with the macro benchmark. This separation prevents market prices from contaminating the definition of economic normality.

The first cross-asset objective should not be “maximize Sharpe.” It should be diagnostic. Ask whether certain benchmark configurations are followed by systematic changes in yields, curves, credit spreads, dollar returns, gold, equity sectors or volatility. Keep release timing and market-pricing expectations explicit. Only after a relationship survives walk-forward testing should it be promoted from an interesting historical association to a usable decision input.

Layer	Question	Example test
Economic benchmark	What is unusual in the real economy?	Walk-forward indicator score and percentile stability
Relationship context	Which macro indicators historically move together or diverge?	Rolling correlation / lead-lag / redundancy stability
Market pricing	What does the market already reflect?	Yield curve, breakeven, spread or valuation state
Cross-asset mapping	What tends to happen after this benchmark configuration?	Forward return distributions by horizon, asset and regime
Decision layer	Is the gap large enough to justify risk?	Out-of-sample return/risk/cost tests

BEFORE AGENT PRODUCTION

17. Validation and Backtest Program

The next research phase should audit the benchmark in the same way the current notebooks audited transformations. The key principle is to separate economic validity from investment usefulness. A

benchmark can be a good descriptive coordinate system even if it has no tradable alpha; conversely, a profitable historical mapping can be spurious if the benchmark leaks future revisions or changes model definitions after seeing the answer.

Stage A - benchmark validation

- Walk-forward reference fitting: every historical score must use only data available before that observation.
- Vintage-aware replay where practical, especially for revised macro releases; store as-of and observation dates separately.
- Calibration stability: compare expanding, rolling, robust and regime-conditioned references without selecting windows solely for downstream returns.
- Outlier policy audit: measure how many observations are excluded from calibration, whether the result is threshold-sensitive, and whether historical shocks remain visible.
- Discrete/regime model audit: inspect sparse-state probabilities, regime occupancy, epoch sensitivity and stability through subperiods.

Stage B - relationship validation

- Rerun correlation, rolling, lead-lag and redundancy on walk-forward slices and compare whether the relationships are stable enough to be useful context.
- Add publication/release timing to lead-lag tests; an observation-time lead is not a tradable information lead.
- Record when correlations reverse and whether those reversals align with documented structural changes or simply temporary crises.

Stage C - agent usefulness audit

- Create dated historical cases where the agent receives only the benchmark packet and period-appropriate evidence.
- Compare benchmark-only interpretation versus benchmark + user dump versus benchmark + external research.
- Score factual grounding, contradiction detection, missing-data identification, hypothesis diversity and later outcome calibration.
- Use ablation: if relationship maps or LLM research add little value, retain the result rather than inventing a benefit.

Stage D - cross-asset tests

- Attach asset returns only after the benchmark/state timestamp is fixed and release timing is respected.
- Evaluate forward-return distributions by horizon; do not optimize the benchmark transformation to the asset return target.
- Test robustness across rates, FX, credit, commodities, gold and equities before claiming a generalized market signal.

ENGINEERING

18. Implementation Path into CHronoZ

The Jupyter notebooks are the laboratory. They should not be the runtime dependency of the agent. Once a benchmark method passes review, its transform, calibration rule and scoring logic should be extracted into small Python modules with deterministic tests. The agent should call one local process or API that returns JSON; it should never execute ad-hoc notebook cells or reimplement formulas in a prompt.

Proposed runtime path

cleaned data / release-aware source

- > transform registry
- > series-specific benchmark model
- > IndicatorState objects
- > relation-view service
- > BenchmarkPacket JSON
- > CHronoZ reasoning agent
- > Macro Journal / Research Outlook
- > future outcome + evaluation record

Because the current repository’s active cleaning logic lives in dataCleansing.py and the existing src/chronoz_quant package is legacy, the productionization step should be treated as a deliberate migration rather than pretending that the source package is already canonical. A sensible next structure is src/chronoz_benchmark/{transforms, models, calibration, relations, packet, backtest}/ with explicit versioned registries.

Minimum agent endpoints

Endpoint / function	Purpose
get_indicator_state(series_id, as_of)	Return one current benchmark state with provenance and model-family metadata.
get_benchmark_packet(as_of, series_ids=None)	Return selected or full indicator states plus warnings.
get_relationship_context(series_ids, as_of)	Return correlation, rolling, lead-lag and redundancy context for a selected set.
get_historical_analogues(query_state)	Later R&D function; retrieve past configurations without claiming a forecast.
record_research_run(...)	Persist the evidence, benchmark version, LLM hypothesis, user edits and future evaluation target.

NEXT WORK

19. Research Roadmap

Milestone	Deliverable	Acceptance idea
B0 Registry	Machine-readable 36-series benchmark registry	Every transform/model/tail policy/version explicit and testable.
B1 Runtime scoring	Python IndicatorState scorer	Matches audited notebook outputs on frozen test cases.
B2 Walk-forward validation	Historical benchmark replay	No look-ahead; reference cutoff visible; stable error/audit reports.
B3 Relationship service	Matrix/rolling/lead-lag/redundancy query layer	Agent can request context for selected series without loading the entire matrix.
B4 BenchmarkPacket	Stable JSON contract	One schema handles z, robust, discrete and regime-conditioned families.
B5 Agent integration	Benchmark-aware CHronoZ workflow	LLM cites numeric state, finds contradictions, and cannot overwrite scores.
B6 Historical case audit	Human-scored research runs	Evaluate reasoning quality and missing-

Milestone	Deliverable	Acceptance idea
		data detection.
B7 Cross-asset mapping	Release-aware forward-return study	No benchmark re-optimization to maximize asset performance.
B8 Pricing integration	Consensus and market-pricing overlays	Reality, expectations and pricing remain separate objects.
B9 Temporal learning	Evaluation records + analogue retrieval	Every run stores what was known, believed and later realized.

TRUST MODEL

20. Epistemic Discipline

Failure mode	v0.3 rule
Look-ahead bias	Reference fitting and relationship tests must use only history available before the scored date.
Revised-data leakage	Use vintage/as-of data when practical; always record release/observation timing.
One-score oversimplification	Keep indicator states and disagreements visible; composites are optional derived views.
Distribution forcing	Use series-specific transforms and model families; do not force Gaussian z-scores onto discrete/regime data.
Outlier deletion	Preserve historical events; exclusion from ordinary calibration must be explicit and auditable.
Correlation = causation	Matrices are research context, not mechanisms.
Lead-lag = forecast	Observation alignment does not prove publication-time predictive power.
Redundancy = delete	Correlated indicators may retain distinct meaning; use redundancy as weighting/context metadata.
Narrative override	LLM cannot rewrite a numeric benchmark because a story is attractive.
Backtest contamination	Do not choose the economic benchmark by maximizing trading Sharpe.
Fake confidence	Do not emit precise causal probabilities without historical calibration.
Hidden model drift	Version transforms, calibration windows, regime definitions and epoch boundaries.

Final principle

Measure the movement. Place it in history. Inspect the relationships. Preserve the disagreement. Then let the researcher ask the next question.

REPOSITORY MAP

APPENDIX A. Notebook Registry

The repository currently contains 44 Jupyter notebooks: eight cross-series / benchmark research notebooks and 36 one-series distribution notebooks. The appendix treats them as research evidence and documentation, not production runtime components.

Notebook	Role
notebook/matrix-corrAnalysis.ipynb	Full-history monthly raw/change Pearson-Spearman matrix, overlap counts and transformation registry.
notebook/rolling-correlation.ipynb	Trailing 60-month Pearson paths, stability summaries and latest-window matrix.
notebook/lead-lag-correlation.ipynb	-12 to +12 month descriptive lag scan with full profiles and overlap controls.
notebook/redundancy-map.ipynb	Strict Pearson/Spearman overlap screen and connection map for candidate redundancy.
notebook/deltaAnalysis/analysis.ipynb	Exploratory 36-series / 71-candidate change-distribution study and suitability ranking.
notebook/deltaAnalysis/audit-PPIACO.ipynb	PPIACO log-delta normalization, empirical ranges, tail preservation and stationarity diagnostics.
notebook/deltaAnalysis/audit-PCEC96.ipynb	PCEC96 outlier-robust calibration, retained shock history and threshold sensitivity.
notebook/deltaAnalysis/audit-nonstandard-delta-models.ipynb	Discrete, robust, regime-conditioned and structural-epoch experiments for six difficult series.

NATIVE DISTRIBUTIONS

APPENDIX B. 36 Individual Data-Learning Notebooks

Category	ID	Variable	Notebook path	Purpose
Growth / Activity	INDPRO	Industrial Production	notebook/data-learning/growth_activity/INDPRO.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	IPMAN	Manufacturing Production	notebook/data-learning/growth_activity/IPMAN.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	PCEC96	Real Personal Consumption Expenditure	notebook/data-learning/growth_activity/PCEC96.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	W875RX1	Real Personal Income excluding transfers	notebook/data-learning/growth_activity/W875RX1.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	RPI	Real Personal Income	notebook/data-learning/growth_activity/RPI.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	RRSFS	Real Retail & Food Services Sales	notebook/data-learning/growth_activity/RRSFS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	HOUST	Housing Starts	notebook/data-learning/growth_activity/HOUST.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Growth / Activity	PERMIT	Building Permits	notebook/data-learning/growth_activity/PERMIT.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	PAYEMS	Nonfarm Payroll Employment	notebook/data-learning/labor/PAYEMS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	UNRATE	Unemployment Rate	notebook/data-learning/labor/UNRATE.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	ICSA	Initial Claims	notebook/data-learning/labor/ICSA.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	AWHAETP	Average Weekly Hours - Total Private	notebook/data-learning/labor/AWHAETP.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	AWHMAN	Average Weekly Hours - Manufacturing	notebook/data-learning/labor/AWHMAN.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	JTSJOL	Job Openings	notebook/data-learning/labor/JTSJOL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	CIVPART	Labor Force Participation Rate	notebook/data-learning/labor/CIVPART.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Labor	CES0500000003	Average Hourly Earnings - Total Private	notebook/data-learning/labor/CES0500000003.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Inflation	CPIAUCSL	CPI All Items	notebook/data-learning/inflation/CPIAUCSL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Inflation	CPILFESL	Core CPI	notebook/data-learning/inflation/CPILFESL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Inflation	PCEPI	PCE Price Index	notebook/data-learning/inflation/PCEPI.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes

Category	ID	Variable	Notebook path	Purpose
Inflation	PCEPILFE	Core PCE Price Index	notebook/data-learning/inflation/PCEPILFE.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Inflation	PPIFID	Producer Prices - Final Demand	notebook/data-learning/inflation/PPIFID.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Inflation	PPIACO	Producer Prices - All Commodities	notebook/data-learning/inflation/PPIACO.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Credit	BUSLOANS	Commercial & Industrial Loans	notebook/data-learning/credit/BUSLOANS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Credit	TOTALSL	Consumer Credit	notebook/data-learning/credit/TOTALSL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Credit	DRTSCILM	SLOOS Tightening C&I	notebook/data-learning/credit/DRTSCILM.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Credit	DRSDCILM	SLOOS Stronger Demand C&I	notebook/data-learning/credit/DRSDCILM.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Credit	DRBLACBS	Business Loan Delinquency	notebook/data-learning/credit/DRBLACBS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Policy / Liquidity	FEDFUNDS	Federal Funds Rate	notebook/data-learning/policy_liquidity/FEDFUNDS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Policy / Liquidity	TOTRESNS	Total Bank Reserves	notebook/data-learning/policy_liquidity/TOTRESNS.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Policy / Liquidity	WRESBAL	Reserve Balances	notebook/data-learning/policy_liquidity/WRESBAL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Policy / Liquidity	WALCL	Federal Reserve Total Assets	notebook/data-learning/policy_liquidity/WALCL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
Policy / Liquidity	M2SL	M2 Money Supply	notebook/data-learning/policy_liquidity/M2SL.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
External / Structural	EXPGSC1	Real Exports	notebook/data-learning/external_structural/EXPGSC1.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
External / Structural	IMPGSC1	Real Imports	notebook/data-learning/external_structural/IMPGSC1.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
External / Structural	OPHNFB	Labor Productivity	notebook/data-learning/external_structural/OPHNFB.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes
External / Structural	TCU	Capacity Utilization	notebook/data-learning/external_structural/TCU.ipynb	Native-frequency level distribution, coverage, histogram, boxplot, audit notes

MODEL REGISTRY

APPENDIX C. Current 36-Series Benchmark Selection

Rank	Series	Category	Preferred change	Model family	Research note
1	HOUST	Growth / Activity	dlog(x)	SELECTED_DELTA	Excellent log-delta historical movement reference
2	PERMIT	Growth / Activity	dlog(x)	SELECTED_DELTA	Cyclical positive quantity; stable

Rank	Series	Category	Preferred change	Model family	Research note
					proportional changes
3	JTSJOL	Labor	dlog(x)	SELECTED_DELTA	Trending level becomes more comparable in growth space
4	OPHNFB	External / Structural	dlog(x)	SELECTED_DELTA	Quarterly productivity growth
5	PPIFID	Inflation	dlog(x)	SELECTED_DELTA	Producer-price proportional movement
6	DRSDCILM	Credit	d level	SELECTED_DELTA	Signed diffusion balance; arithmetic change
7	DRTSCILM	Credit	d level	SELECTED_DELTA	Signed/bounded survey balance
8	IMPGSC1	External / Structural	dlog(x)	SELECTED_DELTA	Quarterly import growth
9	EXPGSC1	External / Structural	dlog(x)	SELECTED_DELTA	Quarterly export growth
10	PCEPILFE	Inflation	dlog(x)	SELECTED_DELTA	Core PCE monthly price movement
11	PCEPI	Inflation	dlog(x)	SELECTED_DELTA	PCE monthly price movement
12	CPIAUCSL	Inflation	dlog(x)	SELECTED_DELTA	Headline CPI monthly movement
13	CPILFESL	Inflation	dlog(x)	SELECTED_DELTA	Core CPI monthly movement; persistence caution
14	DRBLACBS	Credit	d level / pp	SELECTED_DELTA	Rate change in percentage points
15	TCU	External / Structural	d level / pp	SELECTED_DELTA	Capacity-utilization point change
16	CIVPART	Labor	d level / pp	SELECTED_DELTA	Participation-rate point change; small/zero moves
17	FEDFUNDS	Policy / Liquidity	d level / bp	SELECTED_DELTA	Policy-rate change; regime-dependent caution
18	PPIACO	Inflation	100*log ratio	DELTA - REGIME-SENSITIVE	Retain tails; z + empirical percentile; cross-check context
19	PCEC96	Growth / Activity	100*log ratio	OUTLIER-ROBUST DELTA	Exclude robustly flagged events from ordinary calibration only
20	W87SRX1	Growth / Activity	dlog(x)	SELECTED_DELTA	Audit-approved; fiscal/COVID tail caution
21	RRSFS	Growth / Activity	dlog(x)	SELECTED_DELTA	Audit-approved; crisis-tail caution
22	CES0500000003	Labor	dlog(x)	SELECTED_DELTA	Audit-approved; short history/composition caution
23	INDPRO	Growth / Activity	dlog(x)	SELECTED_DELTA	Audit-approved; shutdown/reopening tails
24	IPMAN	Growth / Activity	dlog(x)	SELECTED_DELTA	Audit-approved; keep separate research note
25	TOTALSL	Credit	dlog(x)	SELECTED_DELTA	Audit-approved; structural change caution
26	BUSLOANS	Credit	dlog(x)	SELECTED_DELTA	Audit-approved; policy/credit-regime caution
27	M2SL	Policy / Liquidity	dlog(x)	SELECTED_DELTA	Audit-approved;

Rank	Series	Category	Preferred change	Model family	Research note
					acceleration/regime research note
28	AWHAETP	Labor	d level	DISCRETE_DELTA	Empirical PMF; many exact-zero changes
29	AWHMAN	Labor	d level	SELECTED_DELTA	Audit-approved despite zero share and heavy tails
30	RPI	Growth / Activity	dlog(x)	SELECTED_DELTA	Audit-approved; fiscal-transfer shocks
31	PAYEMS	Labor	dlog(x) / jobs	SELECTED_DELTA	Audit-approved; absolute-jobs alternative remains research item
32	ICSA	Labor	dlog(x)	ROBUST_DELTA	Median/MAD + percentile; retain crisis tails
33	WALCL	Policy / Liquidity	dlog(x)	REGIME_ROBUST_DELTA	Within-speed-regime robust scoring
34	WRESBAL	Policy / Liquidity	dlog(x)	REGIME_ROBUST_DELTA	Regime-conditioned robust liquidity normalization
35	TOTRESNS	Policy / Liquidity	dlog(x)	STRUCTURAL_EPOCH + ROBUST	Segment monetary epochs then score robustly
36	UNRATE	Labor	d level / pp	DISCRETE_PLUS_SHOCK	Discrete PMF plus robust shock detector

RESEARCH EVIDENCE

APPENDIX D. Output Artifact Map

Directory	What it contains	How CHronoZ should use it
outputs/correlation/	Change/raw matrices, pair counts, top pairs, transformation checks, provenance	Full-history relationship context; not final benchmark transformations.
outputs/rolling_correlation/	Rolling Pearson paths, stability summaries, latest matrix	State-dependence / relationship-stability context.
outputs/lead_lag/	All lag correlations, best lags, profile charts, provenance	Timing hypotheses for further release-aware validation.
outputs/redundancy/	Candidate pairs, connection counts, screened matrices	Double-counting warning and optional aggregation constraint.
outputs/distributions/<SERIES>/	Native histograms, boxplots, summary stats, coverage and provenance	Pre-transform diagnostic evidence.
notebook/deltaAnalysis/	Delta study + dedicated audits + current-model notes	Current benchmark-family research source of truth.
data/cleaned/	Canonical date,value cleaned inputs	Quantitative inputs; preserve native frequency and gaps.
data/audit/	Cleaning audit, quality summary, removal/review records	Data-quality provenance and warnings.

REFERENCE

APPENDIX E. Formula & Interpretation Glossary

Object	Representation	Plain meaning
Log delta	$100 * \ln(X_t / X_{(t-1)})$	Approximately percentage movement for small changes; scale-comparable across time.
Arithmetic delta	$X_t - X_{(t-1)}$	Native-unit change; useful for rates, survey balances and discrete values.
Standard z	$(\text{delta} - \text{mean}) / \text{std}$	Distance from the chosen reference center in standard-deviation units.
Robust z	$(\text{delta} - \text{median}) / (1.4826 * \text{MAD})$	Distance using robust center/scale when tails distort mean/std.
Modified z	$0.6745 * (\text{delta} - \text{median}) / \text{MAD}$	Robust outlier diagnostic used in PCEC96 and shock screening.
Empirical percentile	$\text{count}(\text{reference} \leq \text{delta}) / N$	Where the latest movement sits in the actual historical distribution.
Discrete PMF	$P(\text{delta} = x)$	Frequency of an exact recorded movement; useful for rounded/zero-heavy series.
Regime-conditioned score	$\text{score}(\text{delta} \mid \text{regime})$	Historical unusualness relative to a selected statistical

Object	Representation	Plain meaning
		state.
Epoch-conditioned score	score(delta structural epoch)	Historical unusualness within a structural monetary period.
Pearson / Spearman	corr(X,Y)	Linear / rank co-movement; descriptive, not causal.
Lead-lag profile	corr(X_t, Y_(t+k))	Observation alignment across lags; not release-aware predictability.
Redundancy candidate	high Pearson + high Spearman	Potential double counting; not automatic deletion.

PROVENANCE

APPENDIX F. Source & Version Notes

Primary project: <https://github.com/Rizuqura/CHronoZ-economicBenchmarkModelling>

Prior paper: CHronoZ Whitepaper v0.2 - Empirical Economic Benchmark, State, Pricing & Decision Architecture (16 September 2026).

Current repository review and benchmark-selection notes reflect the state inspected on 20 September 2026. The paper intentionally distinguishes repository-computed diagnostics from research selections recorded in CURRENT_MODEL_NOTES.md.

Methodological references retained from v0.2: Federal Reserve Bank of Chicago CFNAI methodology, ALFRED real-time/vintage data, and OECD composite-leading-indicator documentation. These are precedents and data infrastructure references, not templates that CHronoZ is assumed to copy.

Version status

v0.3 is a research architecture and interpretation paper. It does not claim that the benchmark has been fully walk-forward validated, trained for prediction, or integrated into the CHronoZ agent runtime.